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PLANT VARIETY RIGHT AND INTELLECTUAL PROPERTY LAW IN NIGERIA: A REVIEW

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ABSTRACT

Plant breeding is the foundation of agriculture, horticulture, nursery and ornamental sectors. It is most times expensive, time consuming and resource intensive. New plant varieties can easily be stolen, reproduced and sold without permission or fair compensation to the breeders. The Plant variety rights (PVR) and intellectual property (IP) are both legal frameworks that protect agricultural innovations, shaping the dynamics of plant breeding, technology dissemination, and global food security. This research paper examines the historical foundation, relationship between PVR and IP, legal framework governing intellectual property rights (IPR), and PVR protection both international agreements and domestic legislation. It also highlights opportunities in enhancing Plant Variety Rights in Nigeria, Conditions for Plant Variety Protection, the benefit of Plant Variety Right to the breeders and the Nation's Economy.

Keywords: *Plant variety, Intellectual property, Breeders, Plant breeders right.*

INTRODUCTION

Plant variety preservation and intellectual property (IP) rights are crucial for innovation, sustainability, and fair access to genetic resources in the complex field of agriculture. An essential part of this system is Plant Variety Rights (PVR), which give breeders incentives to engage in the creation of novel cultivars and hybrids as well as legal protections (Nath, 2020). PVR regimes, which are a part of larger IP regimes, seek to achieve a careful equilibrium between encouraging innovation and guaranteeing the preservation of genetic variation.

Fundamentally, PVR gives breeders an exclusive authority to commercially propagate, market, and distribute their registered plant varieties. PVR regimes encourage the generation of new varieties that are suited to address changing market needs and environmental issues by providing incentives for research and development. PVR systems also act as a safeguard against the unapproved use or exploitation of breeders' intellectual contributions, creating an atmosphere that is favorable to innovation and business ventures in the agriculture industry.

However, the implementation of PVR regulations raises complex ethical, legal and socioeconomic considerations. Concerns about farmers' rights, benefit-sharing arrangements, and access to genetic resources are major problems that highlight the necessity of a comprehensive and inclusive approach to plant variety sustainability. Establishing a strong and just PVR system is crucial as the world population struggles with the demands of food security, climate resilience, and sustainable development. Intellectual property rights for plant varieties can help create a more resilient, inclusive and sustainable agricultural future by negotiating the complex relationship between innovation and equality. This research work evaluates agricultural progress prior to the implementation of PVP Act's, contrasting it with developments in other African nations and also explains some of the provisions of the Act in relation to innovation, attracting foreign investments, and boost economic growth.

LITERATURE REVIEW

Historical foundation

Farmers have been cultivating and growing plants with desired characteristics and resistance to pests and diseases since the inception of agriculture thousands of years ago. Numerous crop types that are suitable for different environmental factors and societal demands have been developed as a result of this selective breeding technique and been initiated. In the 19th and early 20th centuries, advancements in scientific knowledge and techniques led to the emergence of formal plant breeding programs (Allard, 2023). Plant breeders began systematically crossing different varieties to create new hybrids with improved characteristics. With food security and economic growth depending more and more on agricultural innovation, it became clear that plant breeders' intellectual property needed to be protected. However, because of the distinctive qualities of living things and the drawn-out nature of the breeding process, conventional forms of intellectual property, like patents and copyrights, were sometimes insufficient to protect plant variations.

The first modern plant variety protection laws were introduced in the early 20th century in response to these challenges. One of the earliest examples is the Plant Patent Act of 1930 in the United States, which allowed for the patenting of asexually reproduced plants (Maryland Law Review, 1979). Other countries soon followed suit with their own plant variety protection regimes. Recognizing the importance of harmonizing plant variety protection laws to facilitate international trade in plant germplasm and promote agricultural innovation, efforts were made to establish international agreements in this area. One of the most significant developments was the adoption of the International Union for the Protection of New Plant Varieties Convention (UPOV Convention) in 1961. The UPOV system of plant variety protection came into being with the adoption of the International Convention for the Protection of New Varieties of Plants by a Diplomatic Conference in Paris on December 2, 1961. Plant breeding rights was recognized internationally through the Convention, which provides intellectual property protection for breeders and encourage breeders develop new varieties of plant. Nigeria was not a signatory to the International Union for the Protection of New Varieties of Plants (UPOV) Convention (Okonkwo, et al 2021) which to a large extent impeded the exploitation of such rights. It even deprived plant breeders in Nigeria access to high-quality new varieties of plant from foreign countries and also discourages foreign plant breeders since their new plant variety cannot be protected. These lacunae had surreptitiously deprived Nigeria of innovative opportunities in this sector unlike her counterparts – Uganda (Uganda Plant Variety Act2014), Kenya (Seed and Plant Varieties Act 2012) and South Africa (Plant Breeder's Rights Act 1976).

Meanwhile, the Trade-Related Aspects of Intellectual Property Rights Agreement (TRIPS), which Nigeria is a signatory to, recognizes and provides for the patentability of inventions in all fields of technology and specifically calls for the protection of plant varieties either by patent or by an effective sui generis system or by any combination thereof in line with the provisions of Article 27 of the TRIPS Agreement. Member states of the TRIPS Agreement are obligated to comply with this mandate. Section 1(4) (a) of the Nigeria's Patents and Designs Act, 1970 shares similar provisions of excluding plant varieties from patent protection (Adebola, 2019). Some African countries have opted for the adoption of the International Convention for the Protection of New Varieties of Plants (UPOV Convention) while others have enacted a sui generis plant variety protection law. South Africa is one of the few African countries that had a plant variety protection regime in place before the adoption of the TRIPS Agreement. As a member of both the TRIPS Agreement and the UPOV Convention, South Africa has taken legislative measures to protect plant varieties in addition to the protection of patents.

Comparing plant variety systems in other countries, it's evident that regulatory frameworks and breeder rights are crucial for fully exploiting breeding activities, as new plant varieties require significant investment in skills, labor, resources, and time. Intellectual property (IP) protection is, therefore, necessary to serve as an incentive to plant breeders, for the development of new varieties to contribute to sustainable progress in agriculture.

Relationship between Plant Variety Protection and Traditional Intellectual Property Rights in Nigeria

In Nigeria, the relationship between Plant Variety Protection (PVP) and traditional intellectual property rights (IPR) is evolving, especially with the recent enactment of the Plant Variety Protection Act in 2021 which aimed at: promoting increased staple crop productivity for smallholder farmers in Nigeria and encourage investment in plant breeding and crop variety development; promoting increased mutual accountability in the seed sector; and protecting new varieties of plants. The Act aligns with international standards, such as those set by the International Union for the Protection of New Varieties of Plants (UPOV). PVRs grant exclusive rights to breeders of new plant varieties, allowing them to control the propagation and commercialization of their varieties. This system encourages investment in plant breeding and crop variety development, aiming to boost agricultural productivity and innovation. Traditional Intellectual Property Rights (IPR) on the other hand includes patents, trademarks, copyrights, and industrial designs. These rights protect inventions, brand identities, creative works, and aesthetic designs, respectively (Oriakhogba & Olubiyi, 2023). It stimulates innovation across various industries by ensuring creators and inventors who can reap the benefits of their work. It promotes economic growth and competitiveness. However, traditional IPR systems have not typically covered plant varieties.

In some jurisdictions like the United State and Australia, the subject matter of protection covered by a Traditional IP and by a plant breeder’s right might be the same, namely a plant variety while in several other jurisdictions like European Union and India, the subject matter of protection is different, and, in general, plant varieties are excluded from patentability. As far as the International Union for the Protection of New Plant Varieties Convention (UPOV) is concerned, a member is free to protect plant varieties, in addition to the grant of a breeder’s right, by the grant of other titles, particularly patents. This policy matter is left to the sovereign decision of each member.

Legal Framework for Plant Variety and Intellectual Property

The grant and enforcement of plant variety are governed by international treaties and national laws which have been domesticated. Some of these laws include:

International Treaties

UPOV Convention 1961

The International Union for the Protection of New Varieties of Plants (UPOV) is an intergovernmental organization based in Geneva, Switzerland established in 1961. UPOV provides a sui generis system for the protection of plant varieties and the rights of plant breeders by ensuring that their effort and investments are adequately rewarded. There are several versions of the UPOV Convention, with the latest being UPOV 1991. Some of the Rights Protected by the convention includes: Right to reproduce or multiply the protected variety, conditioning it for propagation, exporting and importing the variety and stocking it for any of these purposes (Article 14 of UPOV); Breeders/researcher can use protected plant varieties to create new varieties (Article 14(5) and15 of UPOV); Breeders have the right to save seeds for the future use to the exclusion of other farmers within a reasonable limit and subject to the safeguarding of the interest of the breeders(Article 15(2) of UPOV).

Member countries agree to adhere to certain standards for protecting plant breeders' rights by granting Intellectual Property Rights to the breeder of new plant variety. UPOV criteria for the protection of plant varieties, includes distinctness, uniformity, and stability (DUS), as well as novelty (Article 5-9 of UPOV). It emphasizes the commercial viability of new plant varieties. Its provides a standard duration of protection is 25 years for trees and vines and 20 years for other plants from the date of grant of the breeders right (Article 10 of UPOV).

TRIPS Agreement 1994

The Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) is administered by the World Trade Organization (WTO). According to the Agreement on Trade-Related Aspect of Intellectual Property Right (1994), the treaty covers a broader range of intellectual property rights, including patents, trademarks, copyrights, and trade secrets, in addition to PVP. TRIPS do not specify detailed criteria for plant variety protection but requires member countries to establish effective systems for the protection of plant varieties.

It allows flexibility for countries to determine their own criteria, which may include patent, or by an effective sui generis system (which could be UPOV standards) or by combination of both systems. The minimum duration of protection for plant varieties, under TRIPS is at least 20 years from the filing date for patents or 15 to 20 years for sui generis systems. TRIPS provide for the exclusive rights of the patent holder or the plant breeder, depending on the type of protection granted. These rights may include the right to prevent others from making, using, selling, or importing the protected variety.

Convention on Biological Diversity and Nagoya Protocol 1992

The Convention on Biological Diversity (CBD) was adopted in 1992 at the Earth Summit in Rio de Janeiro, is a comprehensive international treaty aimed at promoting the conservation of biological diversity; the sustainable use of its components, and the fair and equitable sharing of benefits arising from the utilization of genetic resources. It recognizes the sovereign rights of states over their natural resources, including genetic resources, and emphasizes the principle of access and benefit-sharing. The CBD encourages countries to implement measures to regulate access to genetic resources and ensure that the benefits derived from their utilization are shared fairly and equitably with the countries providing those resources.

The Nagoya Protocol is a supplementary agreement to the CBD, adopted in 2010 and entered into force in 2014. It provides a transparent legal framework for the effective implementation of one of the three objectives of the CBD: the fair and equitable sharing of benefits arising out of the utilization of genetic resources. These agreements are aimed to control the access to genetic resources and biodiversity.

National Legal Framework

Nigerian Agricultural Quarantine Service (NAQS)

The Nigeria Agricultural Quarantine Service (NAQS) prevents the introduction, establishment and spread of exotic pests and diseases of plants and planting material into and outside the country in order to safeguard the nation's agricultural economy. It also promotes sanitary and phytosanitary measures for agricultural export, in order to ensure that the nation's indigenous agricultural pest and diseases are not exported. The service control the regional and international movement of plant variety in order to prevent the introduction and spread of diseases; control of the production and registration of agro pesticides and preventing pesticides or mycotoxin contamination of agricultural commodities; increasing agricultural productivity, by contributing to wealth creation and employment generation, in order to ensure food security and rural development.

The Service also set a defined standards on agricultural quarantine procedure; ensuring compliance with International agreements, protocols, conventions and treaties on agricultural quarantine, including forestry, endangered species, plants variety and planting materials; monitoring the agricultural quarantine Regulations and standards; ensuring that agricultural quarantine projects are carried out in compliance with plant variety and public health regulations; reporting to the competent authority of all plant pests and disease outbreaks(Food and Agriculture Organization of the United Nations, 2018).

National Agricultural Seed Council (NASC)

The National Agricultural Seed Council NASC (2007) is instrumental in Nigeria's seed sector regulation, overseeing the Plant Breeders' Rights Act implementation, promote and stimulate the development of dependable seed industry, facilitating the registration of new plant varieties, protect farmers from the sales of poor quality seeds, facilitate the production and marketing of high quality seeds in Nigeria, provide a legal backing for the official testing certification, sales, importation, exportation and use of seed. NASC is responsible for enforcing PVP rights in Nigeria. This includes monitoring the market for potential infringements and taking legal action against unauthorized use of protected varieties. Breeders can also seek legal remedies through the courts if their rights are violated (National Agricultural Seeds Council Act, 2019).

Plant Variety Protection Act, 2021

The Plant Variety Protection Act (2021) provides legal system that encourages plant breeders to develop new plant varieties, by granting intellectual property over a new plant variety with exclusive rights (Federal Republic of Nigeria Gazette, 2021). In May 2021, Nigeria's President Buhari endorsed the Plant Variety Protection Act [PVP] 2021, giving exclusive intellectual property rights to plant breeders for new plant varieties (Okonkwo, et al, 2021).

Highlights of the Plant Variety Protection Act 2021

Establishment of the Plant Variety Protection Office

Section 5 of the Plant Variety Protection Act 2021 established a Plant Variety Protection Office in Nigeria to grant breeders rights, facilitate transfer and licensing of plant breeder's rights, collaborate with local and international bodies whose functions relate to plant breeding and perform other necessary functions.

Conditions for Plant Variety Protection

The Plant Variety Protection Act 2021 provides for the Registrar to grant protection to plant varieties that are New (novel) which means the plant hasn't been commercialized with breeder's consent within one year before filing it in Nigeria, four years in other territories, and six years for trees and vines in such territories (Section 14 (1) PVP Act 2021); Distinctness, this requires that the plant be easily identifiable and differentiated from other known varieties at the time of filing (Section 15 (1) PVP Act 2021); Uniformity requires a sufficiently uniformed characteristic in the features of the propagated plant variety (Section 16 (1) PVP Act 2021); and Stability after repeated propagation or after a particular cycle of propagation, the relevant characteristics should remain unchanged (Section 16 (1)(b) PVP Act 2021).

Priority Claims

The Plant Variety Protection Act 2021 recognizes the right of priority for a maximum of 12 months to any breeder who has duly filed an application for the protection of a plant variety in one of the members of an international organization dealing with plant breeders rights to which Nigeria is a party and every filed application that fulfills the requirements of the Act will be published by the Registrar in the national gazette or 2 national daily newspapers, which would be open for objection within one month of its publication.

Scope of Plant Breeders' Rights

A plant breeder has exclusive rights to propagate materials of a protected variety. Authorization must be obtained from such a breeder for the production and reproduction, conditioning for propagation; offering for sale; selling or marketing; exporting; importing and stocking in relation to any of the mentioned purposes. Any authorization given by the plant breeder may be subject to conditions and limitations. However, such exclusive rights of a plant breeder cannot impede the doing of the following acts by a third party without authorization: Private and non-commercial purposes; Experimental purposes; and for the purpose of breeding any other variety, except it is essentially derived from an initially protected variety.

Procedure for obtaining Plant Breeder's Right

A breeder submits an application to the Registrar, detailing the plant variety, applicant details, and samples. They must also designate their variety under a unique denomination, which must not affect prior third-party rights, to ensure it is registrable. This unique denomination must not affect any existing rights.

- Where the application satisfies the requirement of the Plant Variety Protection Act 2021, the Registrar will publish a notice of the application in the Federal Gazette or in any two national dailies wide circulation.
- Within a month, a person can submit a written objection to an application within one month of publication, stating the grounds for application, statement of facts, affidavit and other proofs required. The Registrar will notify the applicant within two weeks of the filed objection, and the applicant can reply within two weeks of notice.
- The Registrar will grant Breeder's Rights to the Applicant where the application conforms with the requirements of the Act and that there has not been any objection to the application; or where there has been an objection without grounds for the objection

Grounds for Objection

Objection against the grant of a plant variety must be based on the following grounds: That the Applicant is not entitled to file the application; that the application contains a material misrepresentation; or that the contents of the application do not comply with the provisions of the Plant Variety Protection Act 2021 Act or Regulations.

Duration of Plant Breeder's Rights

Plant breeder's right expires after 20 years from the date of grant except for trees and vines whose breeder's right shall expire 25 years after the date of grant. Grant for the above breeders is extendable for an additional 5 years each, upon notice to the Registrar 6 months before the expiration of the original term. The Plant Variety Protection Act 2021 makes provision for the holder of a plant breeder's right to assign or authorize any activity in relation to a registered plant variety. The assignee shall notify the Registrar of the assignment for record purposes.

Infringement

A breeder right can be protected by both civil and criminal measures stipulated in any written law. An action can also be instituted in court against any person who infringes the breeder right (Section 33 (2) Plant Variety Protection Act 2021).

Benefits of Plant Variety Protection to Plant Breeders

1. The PVP Act grants registered plant breeders Intellectual Property rights, thereby providing incentives for the development of new crop varieties and encouraging more farmers to engage in plant breeding.
2. Breeding entities collaborate with farmers and local institutions during plant variety evaluations, enhancing capacity building and creating employment opportunities.
3. Intellectual Property protection for plant breeders boosts investment in breeding and developing new plant varieties, as investors can now rely on a structured protection system.
4. The PVP Act will encourage non-resident or foreign plant breeder applications, enhancing access to foreign plant varieties by ensuring adequate protection of their varieties.
5. It is expected that the proper implementation of the provisions of the PVP will encourage research and development by breeders, thereby leading to higher agricultural productivity, and a boost in the agricultural sector would have a reverberating effect on Nigeria's economy.

CONCLUSION

The passing of the Plant Protection Variety Act heralded numerous opportunities in the agricultural sector as plant breeders can commercialize their intellectual property by assigning or authorizing any activity in relation to any registered plant variety. This also opens investment opportunities for investors seeking to explore this aspect of agriculture and further creates job opportunities for people skilled in agriculture. This development would serve as a catalyst in the agricultural sector to enable high yield by stakeholders in the sector and to further result in massive contribution to Nigeria's GDP.

RECOMMENDATIONS

1. The Registrar should make effective guidelines for the effective implementation of the Act.

2. The National Agricultural Seeds Council (NASC) should be given adequate resources and trained personnel to effectively manage the PVP system. This will endeavour that NASC can handle

applications, conduct examinations and enforce rights where necessary.

3. A well structured awareness program could be implemented at all level of the society to sensitize farmers, breeders, legislators and the general public on the intellectual property rights and plant variety rights and how they support innovation and food security.

4. Research should be conducted to assess the impact of PVRs on innovation and access to genetic resources.

Regular monitoring of the market for unauthorized use or infringement on plant varieties. Prompt action should be taken to enforce intellectual property rights through legal means such as sending cease-and-desist letters or pursuing litigation.

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